CHAPTER 06

Network Models and Transportation Problems

Decision Making / Game Theory / Systems Analysis

TEXTBOOK	CHAPTER	WEEK
Modern Management Science, 9th ed.	Ch. 6	Week 10

Management Science · Kitae Kim, 2026 Spring, Korea Aerospace University

Made by Kitae Kim, formatted by Codex.

Graphs and Networks

GRAPH = (NODES, EDGES)

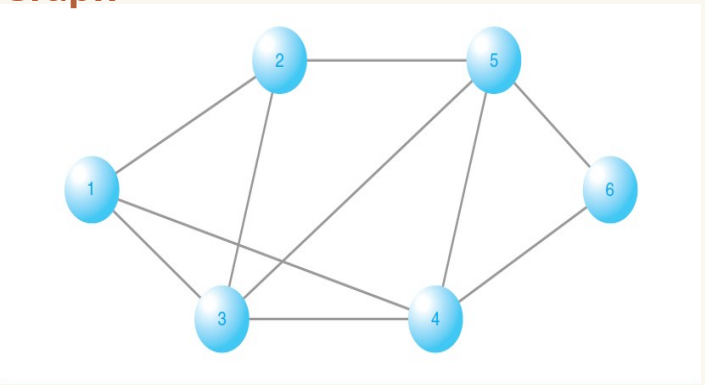
A graph represents the pattern of connections between nodes.

NETWORK = (NODES, EDGES, ATTRIBUTES)

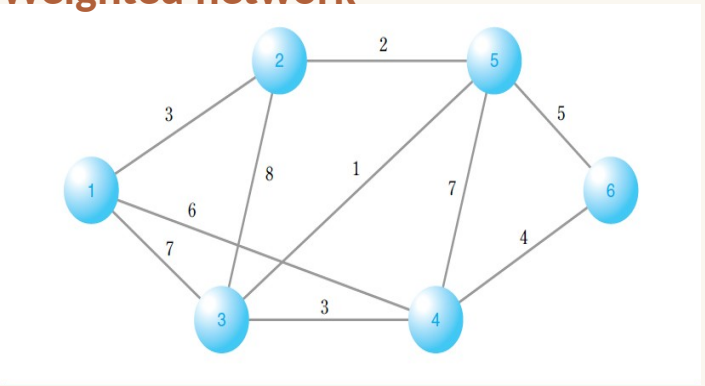
A network adds information about nodes or edges.
It helps us study flows and identify important nodes.

Categorical: gender, occupation, region, date, type
Numerical: strength, cost, distance, time, capacity

Graph



Weighted network



Types of Network Problems

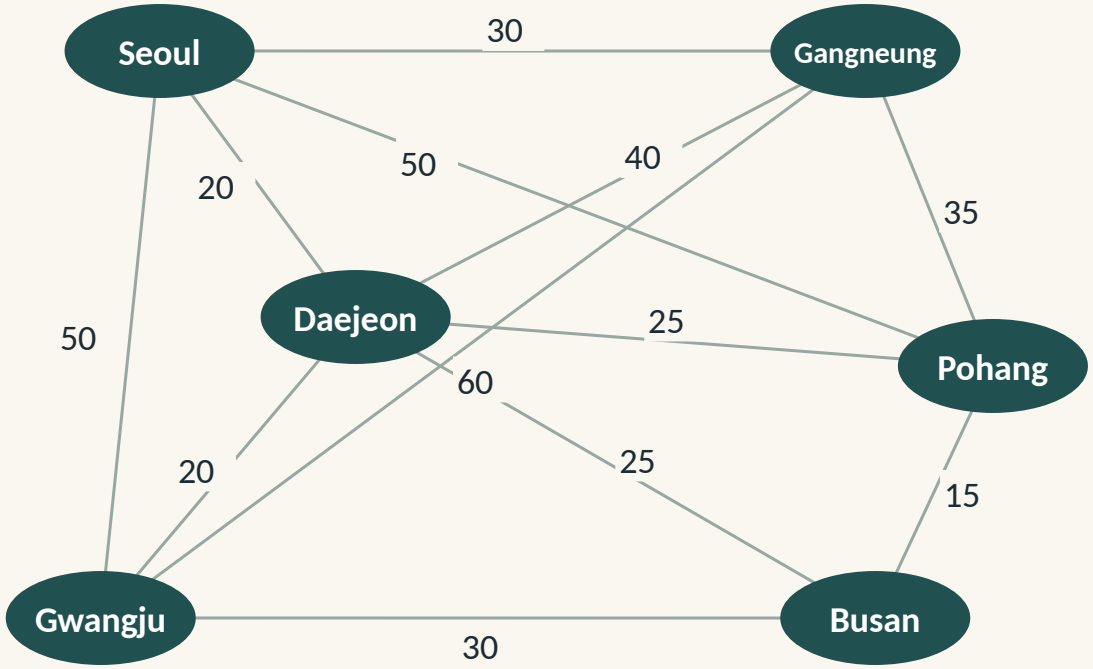
Facility layout / design	Arrange or connect facilities, such as machines or base stations.
Routing	Find the best route between an origin and a destination.
Flow optimization	Determine how much traffic or material to send through a network.
Transportation	Ship directly from supply locations to demand locations.
Transshipment	Ship through intermediate locations or distribution hubs.
Assignment	Assign N workers or machines to N tasks, one to one.

Example 6-1: Connecting Six Offices

Company A has offices in Seoul, Busan, Gwangju, Daejeon, Gangneung, and Pohang.

It plans to build a private data network connecting all six offices. Data may pass through intermediate offices using a store-and-forward system.

Find a set of links that connects every office at minimum total cost.



Link costs are in units of KRW 100 million

Example 6-1: Minimum Spanning Tree

Minimal Spanning Tree

1. Start with any node.

2. Add the lowest-cost edge connecting a node in the current tree to a node outside it.

3. Repeat until all nodes are connected.

Do not add an edge that creates a cycle.

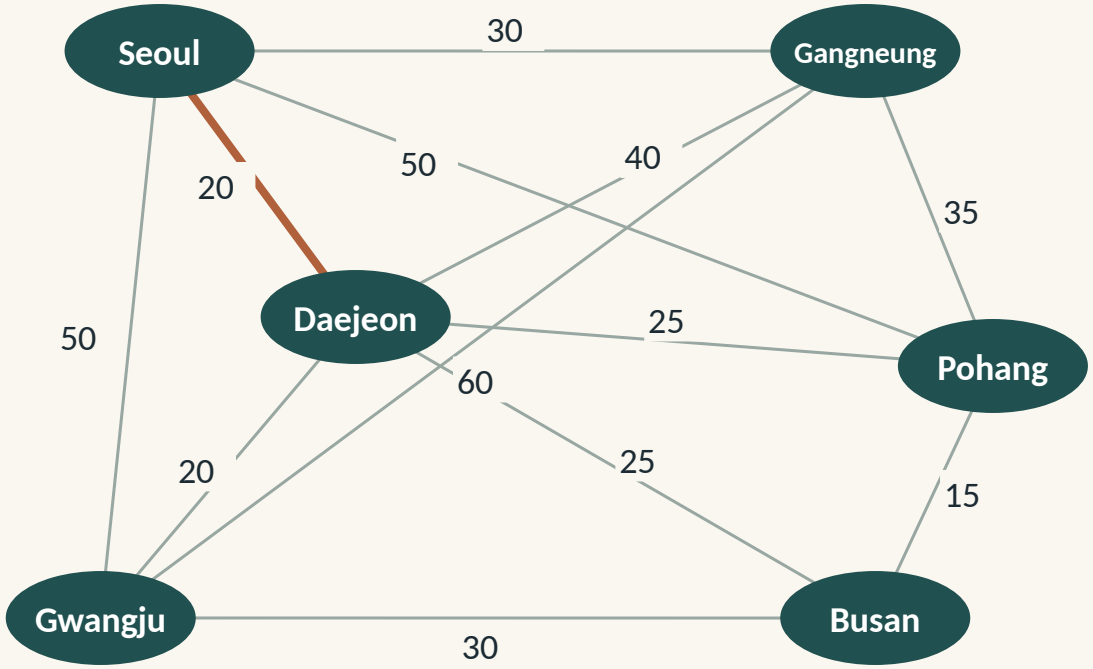


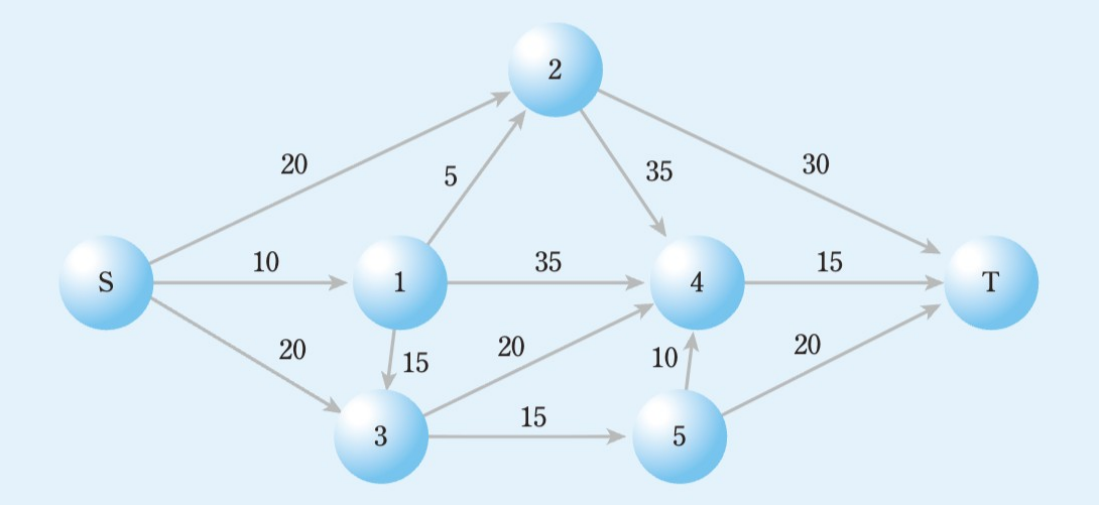
Illustration: start with Seoul → Daejeon (20).

Example 6-2: Shortest Path Problem

A commuter travels from home (S) to work (T). The number on each directed arc is the travel time in minutes.

Possible routes include S → 1 → 4 → T and S → 3 → 5 → T.

Formulate an integer programming model to find the shortest route.

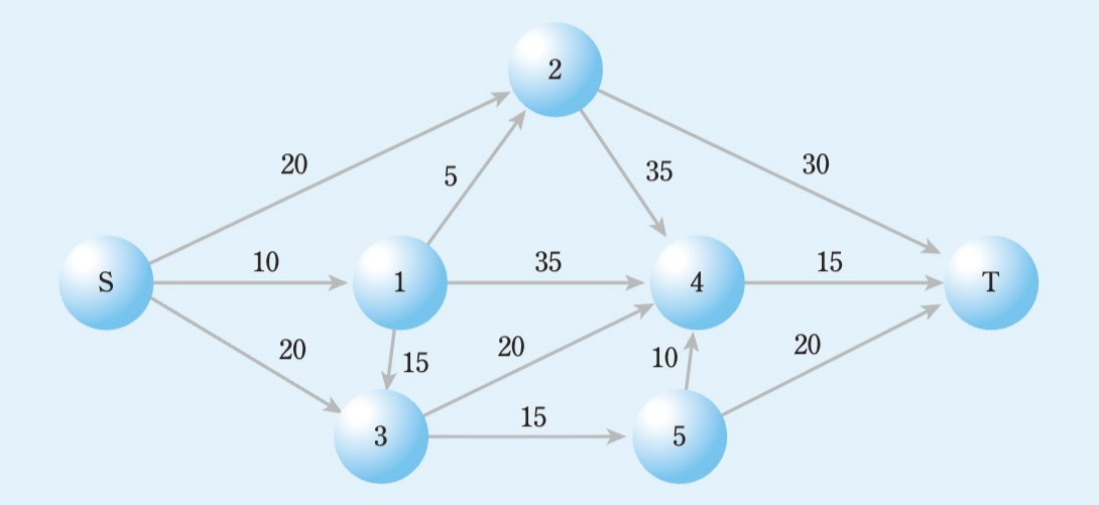


Example 6-2: Shortest-Path Model

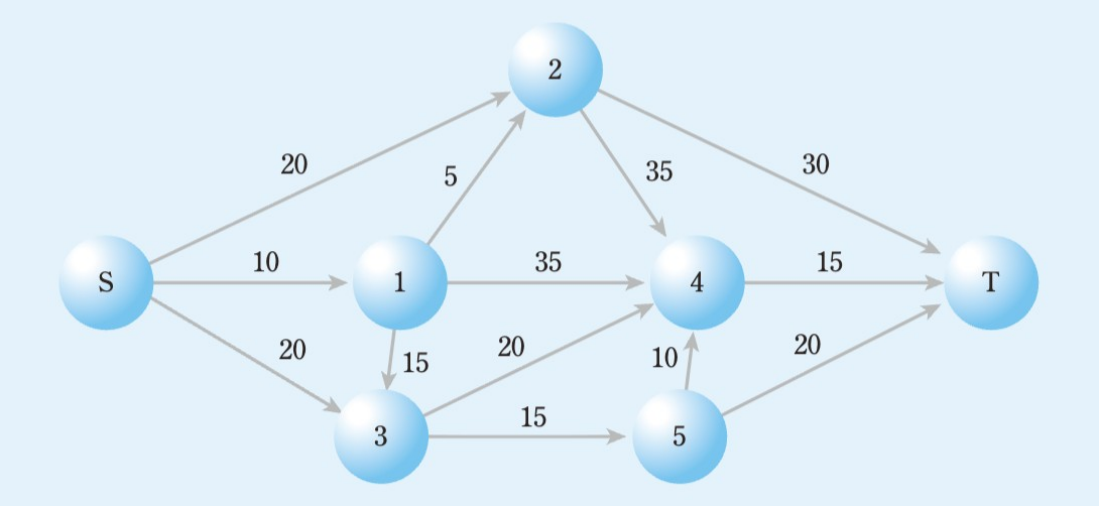
VARIABLES AND OBJECTIVE

$$x_{ij} = \begin{cases} 1, & if\ i \rightarrow j\ is\ used \\ 0, & otherwise. \end{cases}$$

Minimize $10x_{S1} + 20x_{S2} + 20x_{S3}$
$+ 5x_{12} + 15x_{13} + 35x_{14}$
$+ 35x_{24} + 30x_{2T} + 20x_{34}$
$+ 15x_{35} + 10x_{54}$
$+ 15x_{4T} + 20x_{5T}$



Example 6-2: Shortest-Path Model



FLOW-BALANCE CONSTRAINTS

$$x_{S1} = x_{12} + x_{13} + x_{14}$$
$$x_{S2} + x_{12} = x_{24} + x_{2T}$$
$$x_{S3} + x_{13} = x_{34} + x_{35}$$
$$x_{14} + x_{24} + x_{34} + x_{54} = x_{4T}$$
$$x_{35} = x_{54} + x_{5T}$$
$$x_{S1} + x_{S2} + x_{S3} = 1$$
$$x_{2T} + x_{4T} + x_{5T} = 1$$

$$x_{ij} \in \{0,1\}$$

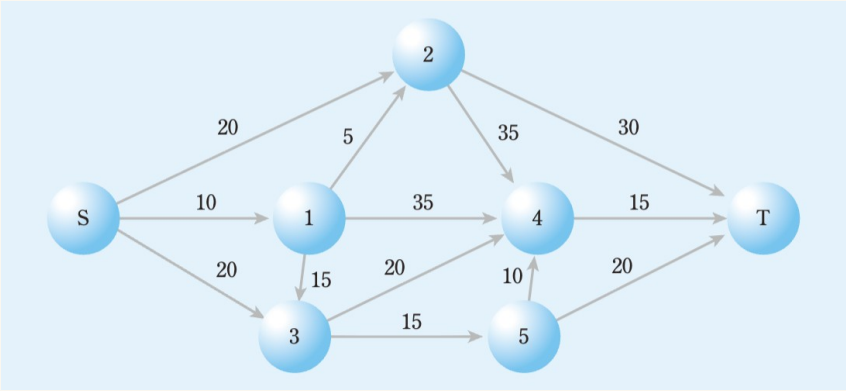
An algorithmic alternative: Dijkstra's algorithm.

Example 6-2: Shortest-Path Model

VARIABLES AND OBJECTIVE

$$x_{ij} = \begin{cases} 1, & if\ i \rightarrow j\ is\ used \\ 0, & otherwise. \end{cases}$$

Minimize $10x_{S1} + 20x_{S2} + 20x_{S3} + 5x_{12} + 15x_{13} + 35x_{14} + 35x_{24} + 30x_{2T} + 20x_{34} + 15x_{35} + 10x_{54} + 15x_{4T} + 20x_{5T}$



FLOW-BALANCE CONSTRAINTS

$x_{S1} = x_{12} + x_{13} + x_{14}$

$x_{S2} + x_{12} = x_{24} + x_{2T}$

$x_{S3} + x_{13} = x_{34} + x_{35}$

$x_{14} + x_{24} + x_{34} + x_{54} = x_{4T}$

$x_{35} = x_{54} + x_{5T}$

$x_{S1} + x_{S2} + x_{S3} = 1$

$x_{2T} + x_{4T} + x_{5T} = 1$

$x_{ij} \in \{0,1\}$ for every available arc.

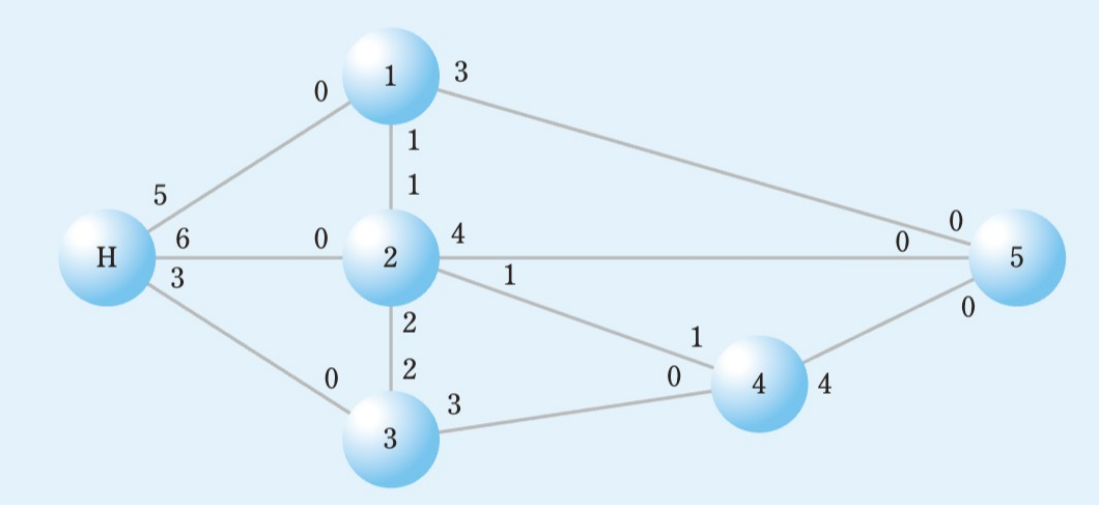
An algorithmic alternative: Dijkstra's algorithm.

Example 6-3: Maximum Flow Problem

A dam (H) supplies water to cities 1–5 through a pipeline network. An earthquake has destroyed the water-storage facility in city 5.

Determine the maximum amount of water that can be sent from the dam to city 5.

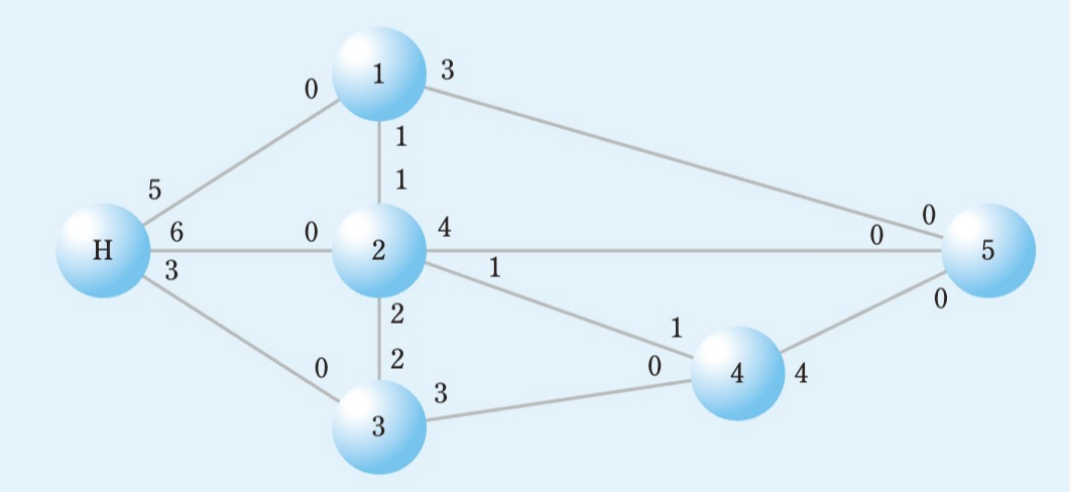
The numbers indicate capacities per unit of time; a capacity of 0 means that flow is not allowed in that direction.



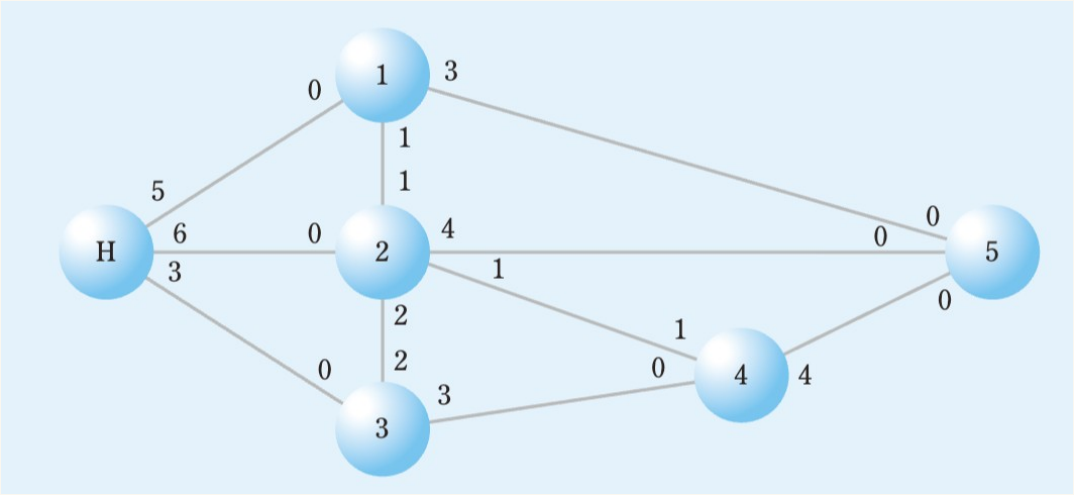
Example 6-3: Maximum-Flow Model

VARIABLES AND OBJECTIVE

- x_{ij} = water sent from node i to node j per unit of time.
- Maximize $x_{15} + x_{25} + x_{45}$



Example 6-3: Maximum-Flow Model



FLOW BALANCE AT INTERMEDIATE CITIES

$x_{H1} + x_{21} = x_{12} + x_{15}$

$x_{H2} + x_{12} + x_{32} + x_{42} = x_{21} + x_{23} + x_{24} + x_{25}$

$x_{H3} + x_{23} = x_{32} + x_{34}$

$x_{24} + x_{34} = x_{42} + x_{45}$

CAPACITY CONSTRAINTS

$x_{H1} \le 5;\ x_{H2} \le 6;\ x_{H3} \le 3$

$x_{12} \le 1;\ x_{21} \le 1$

$x_{23} \le 2;\ x_{32} \le 2$

$x_{15} \le 3;\ x_{25} \le 4$

$x_{24} \le 1;\ x_{42} \le 1$

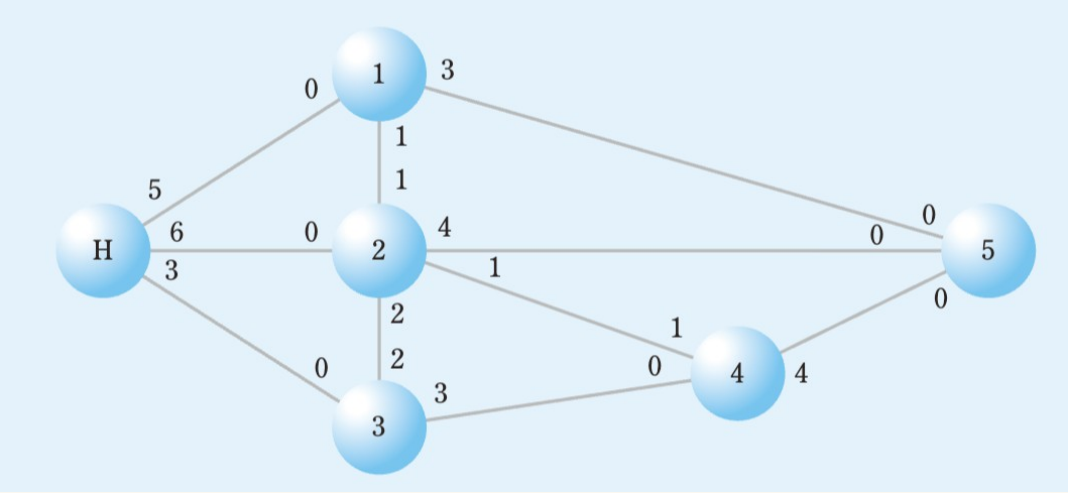
$x_{34} \le 3;\ x_{45} \le 4$

$x_{ij} \ge 0$ for all allowed arcs. All zero-capacity directions carry no flow.

Example 6-3: Maximum-Flow Model

VARIABLES AND OBJECTIVE

- x_{ij} = water sent from node i to node j per unit of time.
- Maximize $x_{15} + x_{25} + x_{45}$



FLOW BALANCE AT INTERMEDIATE CITIES

$$x_{H1} + x_{21} = x_{12} + x_{15}$$

$$x_{H2} + x_{12} + x_{32} + x_{42} = x_{21} + x_{23} + x_{24} + x_{25}$$

$$x_{H3} + x_{23} = x_{32} + x_{34}$$

$$x_{24} + x_{34} = x_{42} + x_{45}$$

CAPACITY CONSTRAINTS

$x_{H1} \le 5$; $x_{H2} \le 6$; $x_{H3} \le 3$

$x_{12} \le 1$; $x_{21} \le 1$

$x_{23} \le 2$; $x_{32} \le 2$

$x_{15} \le 3$; $x_{25} \le 4$

$x_{24} \le 1$; $x_{42} \le 1$

$x_{34} \le 3$; $x_{45} \le 4$

$x_{ij} \ge 0$ for all allowed arcs. All zero-capacity directions carry no flow.

Example 6-4: Transportation Problem

Samsung Cider operates plants in Masan, Daejeon, and Gunsan and warehouses in Busan, Daegu, Gwangju, and Seoul.
Find the shipping plan that meets demand at minimum transportation cost.
Assume that supply must exactly meet demand.

Plant	Busan	Daegu	Gwangju	Seoul	Supply
Masan	8	9	11	16	50
Daejeon	12	7	5	8	80
Gunsan	14	10	6	7	120
Demand	90	70	40	50	250

Unit shipping costs are shown in KRW millions. Supply and demand are in units.

Example 6-4: Formulate the Model

Plant	Busan	Daegu	Gwangju	Seoul	Supply
Masan	8	9	11	16	50
Daejeon	12	7	5	8	80
Gunsan	14	10	6	7	120
Demand	90	70	40	50	250

Balanced transportation model:
Total supply = total demand = 250 units.

Decision variables

- x_{ij} = units shipped from plant i to warehouse j.
- i = 1: Masan, 2: Daejeon, 3: Gunsan;
- j = 1: Busan, 2: Daegu, 3: Gwangju, 4: Seoul.

Objective function

$$\text{Minimize } 8x_{11} + 9x_{12} + 11x_{13} + 16x_{14} + 12x_{21} + 7x_{22} + 5x_{23} + 8x_{24} + 14x_{31} + 10x_{32} + 6x_{33} + 7x_{34}$$

Example 6-4: Formulate the Model

Plant	Busan	Daegu	Gwangju	Seoul	Supply
Masan	8	9	11	16	50
Daejeon	12	7	5	8	80
Gunsan	14	10	6	7	120
Demand	90	70	40	50	250

Balanced transportation model:
Total supply = total demand = 250 units.

Constraints

SUPPLY

$\Sigma_j x_{1j} = 50$

$\Sigma_j x_{2j} = 80$

$\Sigma_j x_{3j} = 120$

DEMAND

$\Sigma_i x_{i1} = 90; \quad \Sigma_i x_{i2} = 70$

$\Sigma_i x_{i3} = 40; \quad \Sigma_i x_{i4} = 50$

$x_{ij} \geq 0$ for all i and j.

Example 6-4: Transportation Model

DECISION VARIABLES

- x_{ij} = units shipped from plant i to warehouse j.
- i = 1: Masan, 2: Daejeon, 3: Gunsan;
- j = 1: Busan, 2: Daegu, 3: Gwangju, 4: Seoul.

MINIMIZE TOTAL TRANSPORTATION COST

$$\text{Minimize } 8x_{11} + 9x_{12} + 11x_{13} + 16x_{14} + 12x_{21} + 7x_{22} + 5x_{23} + 8x_{24} + 14x_{31} + 10x_{32} + 6x_{33} + 7x_{34}$$

Constraints

SUPPLY

$$\Sigma_j x_{1j} = 50$$

$$\Sigma_j x_{2j} = 80$$

$$\Sigma_j x_{3j} = 120$$

DEMAND

$$\Sigma_i x_{i1} = 90; \quad \Sigma_i x_{i2} = 70$$

$$\Sigma_i x_{i3} = 40; \quad \Sigma_i x_{i4} = 50$$

$$x_{ij} \geq 0 \text{ for all i and j.}$$

Example 6-5: Transshipment Problem

R Electronics ships from plants in Denver and Atlanta through hubs in Kansas City and Louisville to four destinations. Find the least-cost shipping plan.

PLANT → HUB ($ PER UNIT)

Plant	Kansas City	Louisville	Supply
Denver	200	300	6000
Atlanta	300	100	4000

HUB → DESTINATION ($ PER UNIT)

Hub	Detroit	Miami	Dallas	New Orleans
Kansas City	200	600	300	600
Louisville	400	400	600	500
Demand	2000	1500	3500	3000

Example 6-5: Formulate the Model

PLANT → HUB ($ PER UNIT)

Plant	Kansas City	Louisville	Supply
Denver	200	300	6000
Atlanta	300	100	4000

HUB → DESTINATION ($ PER UNIT)

Hub	Detroit	Miami	Dallas	New Orleans
Kansas City	200	600	300	600
Louisville	400	400	600	500
Demand	2000	1500	3500	3000

Decision variables

- X_{ik}: units from plant i to hub k.
- Y_{kj}: units from hub k to destination j.

Objective function

Minimize Z

$$= 200X_{11} + 300X_{12} + 300X_{21} + 100X_{22} + 200Y_{11} + 600Y_{12} + 300Y_{13} + 600Y_{14} + 400Y_{21} + 400Y_{22} + 600Y_{23} + 500Y_{24}$$

Example 6-5: Formulate the Model

PLANT → HUB ($ PER UNIT)

Plant	Kansas City	Louisville	Supply
Denver	200	300	6000
Atlanta	300	100	4000

HUB → DESTINATION ($ PER UNIT)

Hub	Detroit	Miami	Dallas	New Orleans
Kansas City	200	600	300	600
Louisville	400	400	600	500
Demand	2000	1500	3500	3000

Constraints

SUPPLY AND HUB FLOW BALANCE

$$X_{11} + X_{12} = 6{,}000$$

$$X_{21} + X_{22} = 4{,}000$$

$$X_{11} + X_{21} = \Sigma_j Y_{1j}$$

$$X_{12} + X_{22} = \Sigma_j Y_{2j}$$

DEMAND

$$Y_{11} + Y_{21} = 2{,}000$$

$$Y_{12} + Y_{22} = 1{,}500$$

$$Y_{13} + Y_{23} = 3{,}500$$

$$Y_{14} + Y_{24} = 3{,}000$$

All $X_{ik}, Y_{kj} \geq 0$.

Example 6-5: Formulate the Model

Decision variables

- X_{ik}: units from plant i to hub k.
- Y_{kj}: units from hub k to destination j.

Objective function

Minimize Z

$$= 200X_{11} + 300X_{12} + 300X_{21} + 100X_{22} + 200Y_{11} + 600Y_{12} + 300Y_{13} + 600Y_{14} + 400Y_{21} + 400Y_{22} + 600Y_{23} + 500Y_{24}$$

Constraints

SUPPLY AND HUB FLOW BALANCE

$$X_{11} + X_{12} = 6{,}000$$

$$X_{21} + X_{22} = 4{,}000$$

$$X_{11} + X_{21} = \Sigma_j Y_{1j}$$

$$X_{12} + X_{22} = \Sigma_j Y_{2j}$$

DEMAND

$$Y_{11} + Y_{21} = 2{,}000$$

$$Y_{12} + Y_{22} = 1{,}500$$

$$Y_{13} + Y_{23} = 3{,}500$$

$$Y_{14} + Y_{24} = 3{,}000$$

All $X_{ik}, Y_{kj} \geq 0$.

Summary

Network vs. Graph

- Graphs describe connections.
- Networks add node and edge attributes.

SELECT A NETWORK MODEL

- Minimum spanning tree
- Shortest path
- Maximum flow
- Transportation
- Transshipment
- Assignment